TRAFFIC IMAGE SEGMENTATION ON STM32M4 & MAX78000

MACHINE LEARNING ON MICROCONTROLLERS

ETH ZURICH, JAN 2024

PHANI JAYANTH JONNALAGADDA

VARSHA JAYAPRAKASH

AGENDA

- BACKGROUND
- THE DATASET
- RELEVANT WORK
- PROJECT GOALS
- THE MODELS
- RESULTS
- DEMO
- BEYOND THE DATASET

BACKGROUND

- Image Segmentation is a Computer Vision task that involves dividing an image into semantically significant segments based on certain criteria.
- Image Segmentation is critical for various applications like autonomous driving, medical imaging, satellite image analysis, computer-aided design, and more.

Focus
- Semantic Segmentation of Traffic Image Data
- Applications in Autonomous Driving

SOTA
- CNN-based Architectures
- U-Net, DeepLab, FCN, SegNet, etc.

Challenges
- Model Complextity for MCUs
- Numerous Segmentation Classes

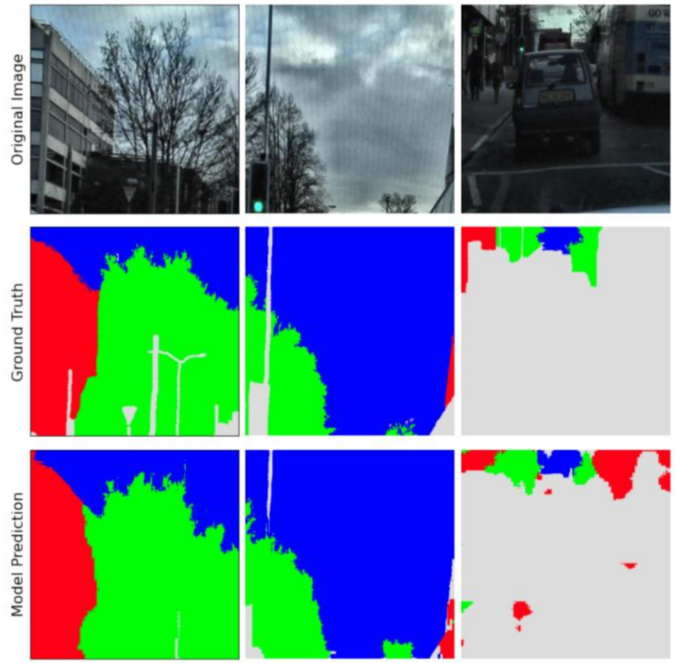
THE DATASET: CITYSCAPES

- **Cityscapes:** The dataset focuses on semantic understanding of urban street scenes.
- **Features:** 30 Classes, 50 cities, and 5000 annotated images.
- **Images:** 1024 x 2048 PNG images.
- **Limitations:**
  - Images are captured only in daylight.
  - Images are captured only in good weather conditions.

RELEVANT WORK

L^3U-Net: Low-Latency Lightweight U-net Based Image Segmentation Model for Parallel CNN Processors

- Published in 2022 by Analog Devices Inc.
- **U-Net-based** tiny image segmentation model for edge devices.
- Trained on *CamVid* and *AISegment* Datasets.
- Tested & Deployed on MAX78000.
- Proposes ***data folding*** technique to reduce latency and enable processing on high resolution input images.



https://arxiv.org/pdf/2203.16528

PROJECT GOALS

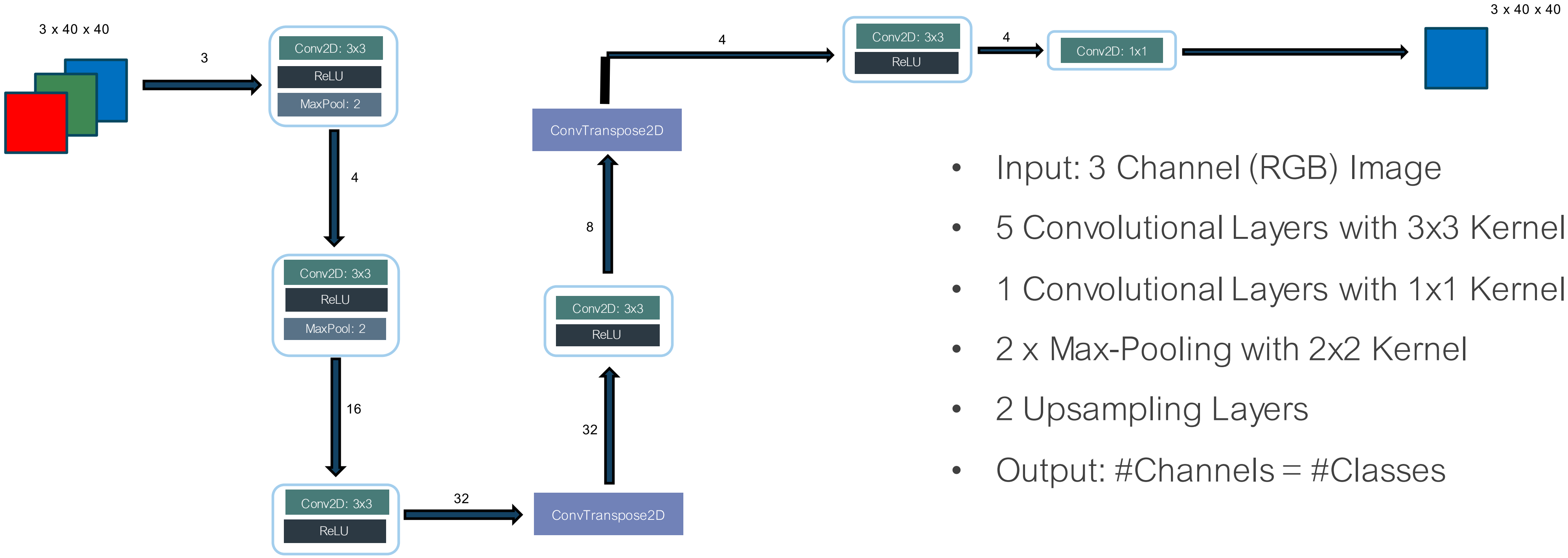
STM32M4

- Implementing a simple & lightweight Encoder-Decoder architecture
- Making the model larger
- Improving the model accuracy
- Further Optimizations using Pruning and Quantization.
- Inference

MAX78000

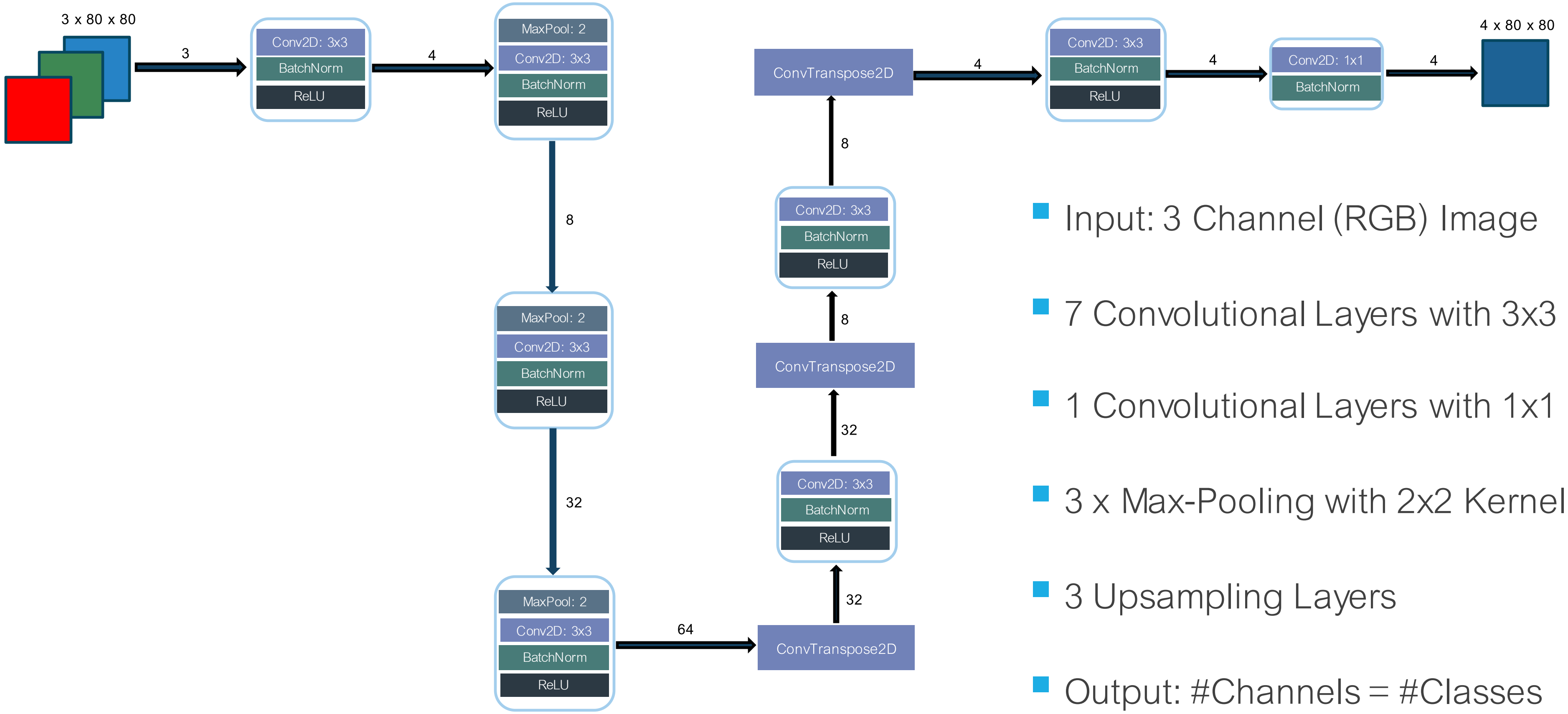
- Implementing the larger U-Net model on MAX78000
- Leveraging the existing U-Net architectures, implemented by Analog Devices Inc.
- Quantization-Aware Training
- Comparision across various architectures
- Inference

THE MODEL: A SIMPLE ENCODER-DECODER NETWORK



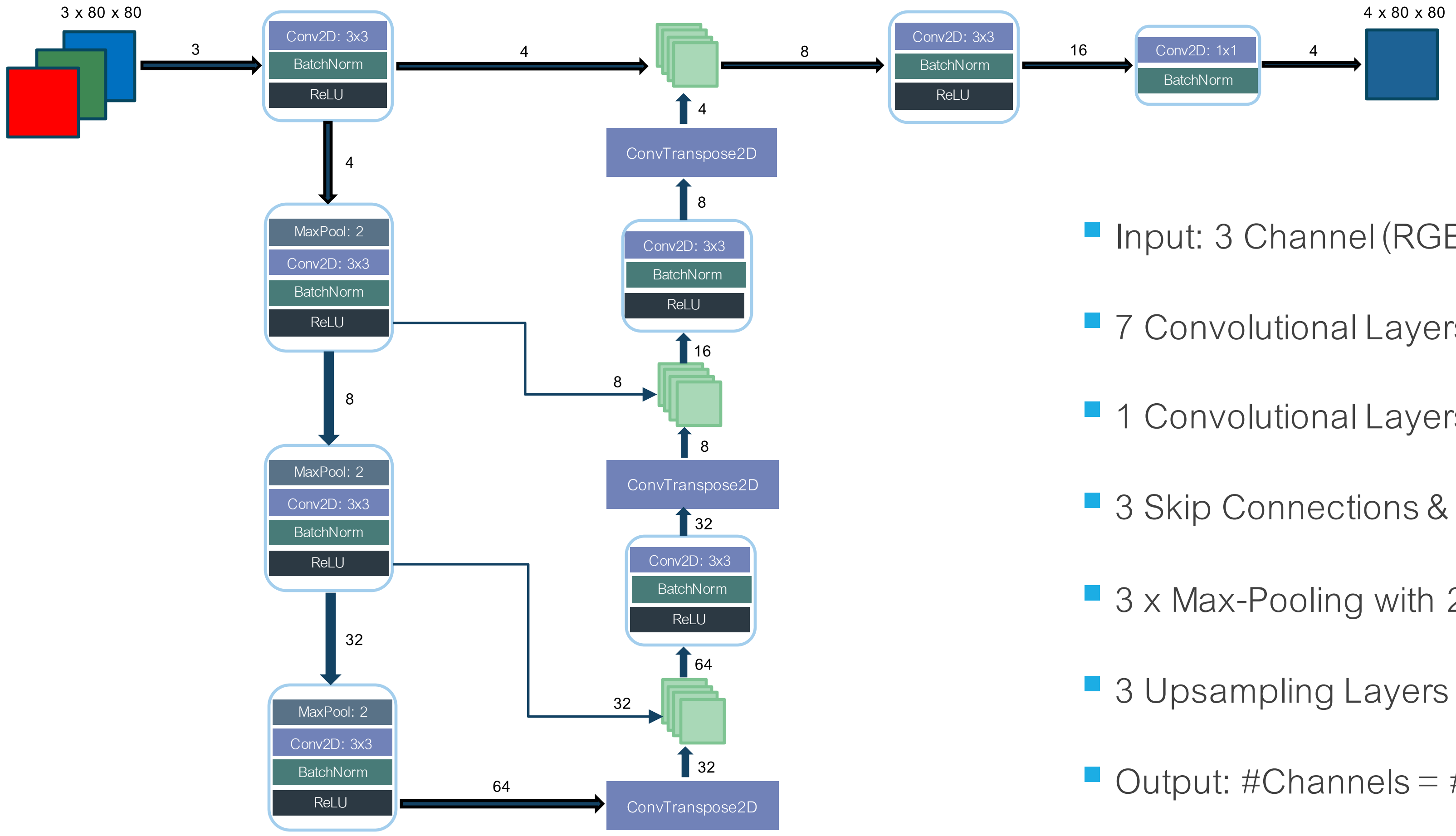
- Input: 3 Channel (RGB) Image
- 5 Convolutional Layers with 3x3 Kernels
- 1 Convolutional Layers with 1x1 Kernel
- 2 x Max-Pooling with 2x2 Kernel
- 2 Upsampling Layers
- Output: #Channels = #Classes

THE MODEL: ENHANCED ENCODER-DECODER NETWORK



- Input: 3 Channel (RGB) Image
- 7 Convolutional Layers with 3x3 Kernels
- 1 Convolutional Layers with 1x1 Kernel
- 3 x Max-Pooling with 2x2 Kernel
- 3 Upsampling Layers
- Output: #Channels = #Classes

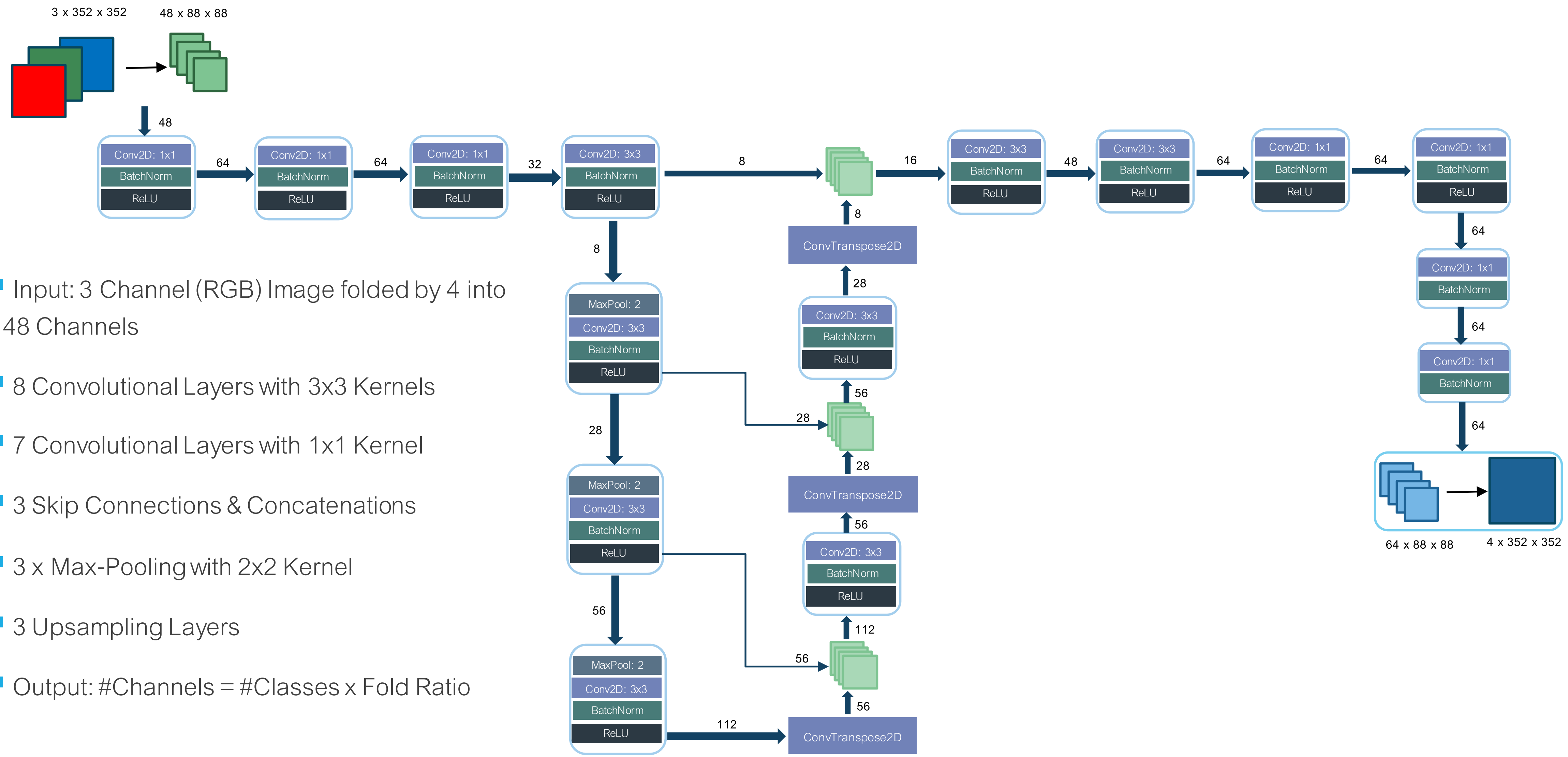
THE MODEL: SMALL U-NET BY ANALOG DEVICES INC.



- Input: 3 Channel (RGB) Image
- 7 Convolutional Layers with 3x3 Kernels
- 1 Convolutional Layers with 1x1 Kernel
- 3 Skip Connections & Concatenations
- 3 x Max-Pooling with 2x2 Kernel
- 3 Upsampling Layers
- Output: #Channels = #Classes

THE MODEL: LARGE U-NET BY ANALOG DEVICES INC.

- Input: 3 Channel (RGB) Image folded by 4 into 48 Channels
- 8 Convolutional Layers with 3x3 Kernels
- 7 Convolutional Layers with 1x1 Kernel
- 3 Skip Connections & Concatenations
- 3 x Max-Pooling with 2x2 Kernel
- 3 Upsampling Layers
- Output: #Channels = #Classes x Fold Ratio



Variation in Number of Semantic Segmentation Classes

Three Classes

STM32M4

Classes:

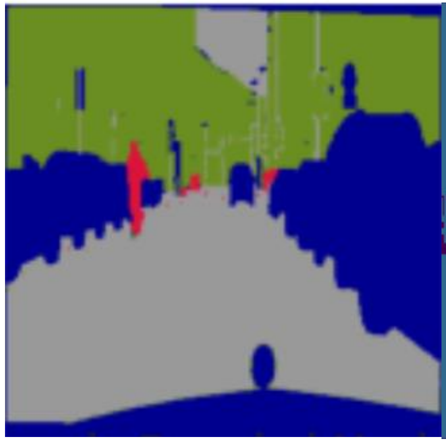
Vehicle, Vegetation, and Void

MAX78000

Classes:

Vehicle, Vegetation, and Void

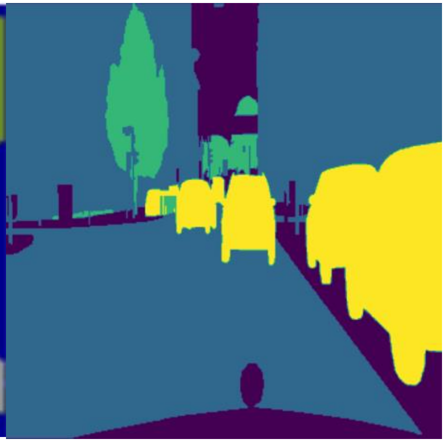
Four Classes



STM32M4

Classes:

Vehicle, Vegetation, Human, and Void



MAX78000

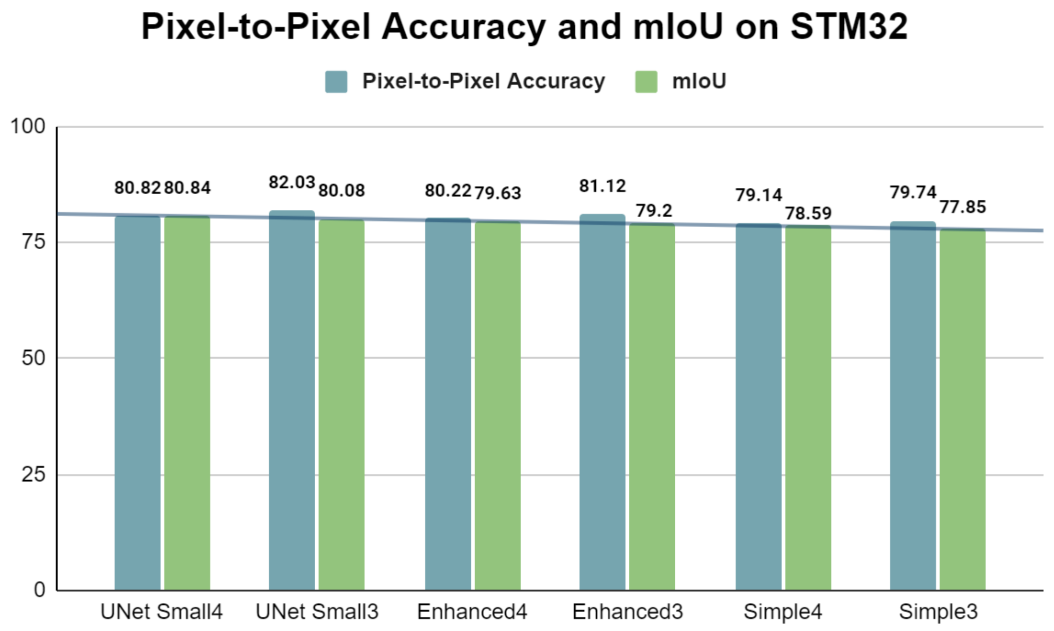
Classes:

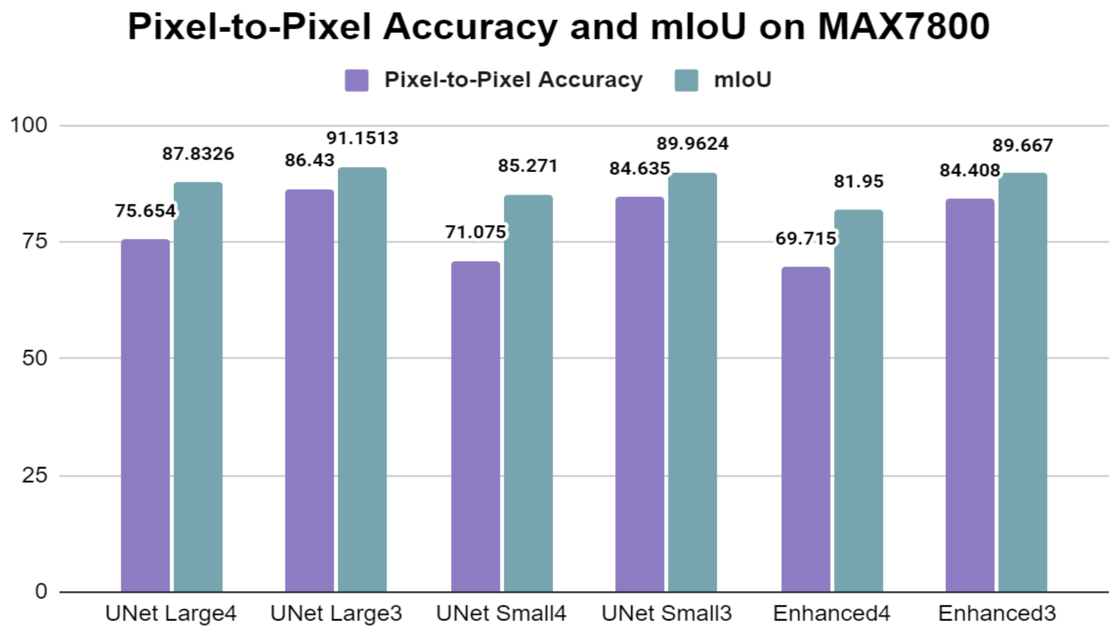
Vehicle, Vegetation, Building, and Void

Data Augmentations

- RandomHorizontalFlip
- RandomAutoContrast
- RandomGrayscale
- ColorJitter
- GaussianBlur
- RandomInvert
- RandomSolarize

Model Performance: Pixel-to-Pixel Accuracy and mIoU





The results observed on STM32 use Unstructured Pruning with Dynamic Sparsity + QAT (float32). The MAX78000 results are obtain ed using QAT with 8 weight bits.

The Pixel-to-Pixel Accuracy and mIoU on *CamVid* across 4 classes reported by Analog Devices Inc. are 91.05% and 84.24% respectively with an inference time of 95.1 ms.

Model Benchmarking (3 Classes)

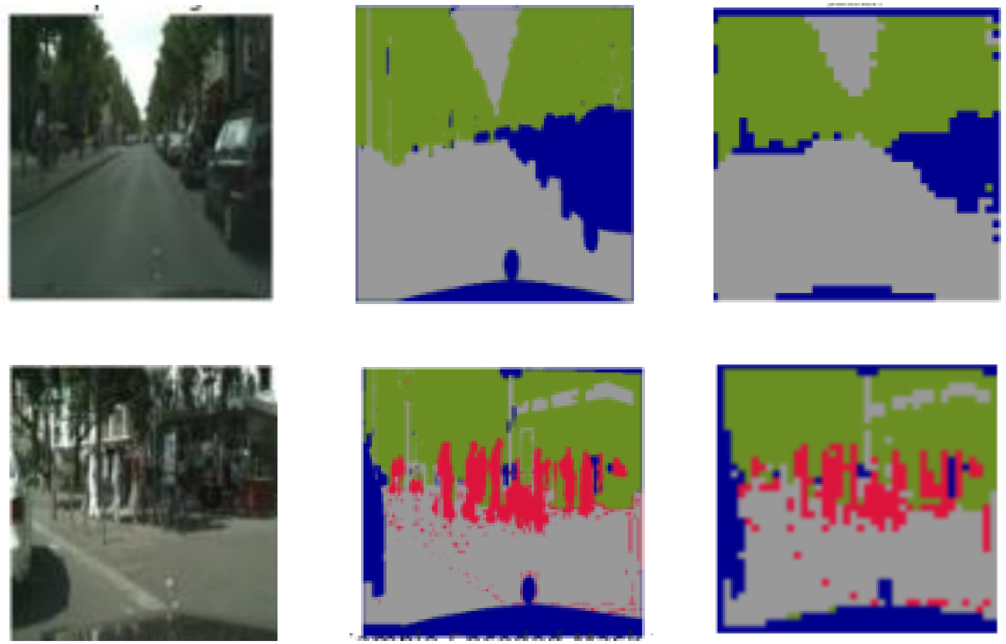
METRIC	SIMPLE	ENHANCED NETWORK		SMALL UNET		LARGE UNET
TOTAL # OF PARAMETERS	STM32M4	STM32M4	MAX78000	STM32M4	MAX78000	MAX78000
	17,323	52,259	52,104	62,487	62,940	277,152
MACs (M)	2.167	5.712	22.426	4.56	34.125	619.769
QUANTIZED MEMORY USAGE	RAM: 66.03 KB Flash: 58.67 KB	RAM: 69.06 KB Flash: 97.92 KB	51.08 KB	RAM: 55.26 KB Flash: 109.89 KB	62.67 KB	274.59 KB
# OF CYCLES (M)	15.68	17.36	0.261	10.48	0.365	4.638
INFERENCE TIME (ms)	196	217	5.216	131	7.3	92.753
THROUGHPUT (MACs/Cycle)	0.138	0.329	85.989	0.432	93.486	133.638

Model Benchmarking (4 Classes)

METRIC	SIMPLE	ENHANCED NETWORK		SMALL UNET		LARGE UNET
	STM32M4	STM32M4	MAX78000	STM32M4	MAX78000	MAX78000
TOTAL # OF PARAMETERS	17,328	52,290	52,108	62,847	62,956	278176
MACs (M)	2.18	5.73	22.451	4.56	34.163	627.697
QUANTIZED MEMORY USAGE	RAM: 66.03 KB Flash: 58.67 KB	RAM: 69.06 KB Flash: 97.92 KB	51.08 KB	RAM: 55.26 KB Flash: 109.89 KB	62.67 KB	275.60 KB
# OF CYCLES (M)	15.6	17.04	0.267	10.96	0.374	4.762
INFERENCE TIME (ms)	195	213	5.348	137	7.485	95.234
THROUGHPUT (MACs/Cycle)	0.139	0.329	84.086	0.438	91.344	131.814

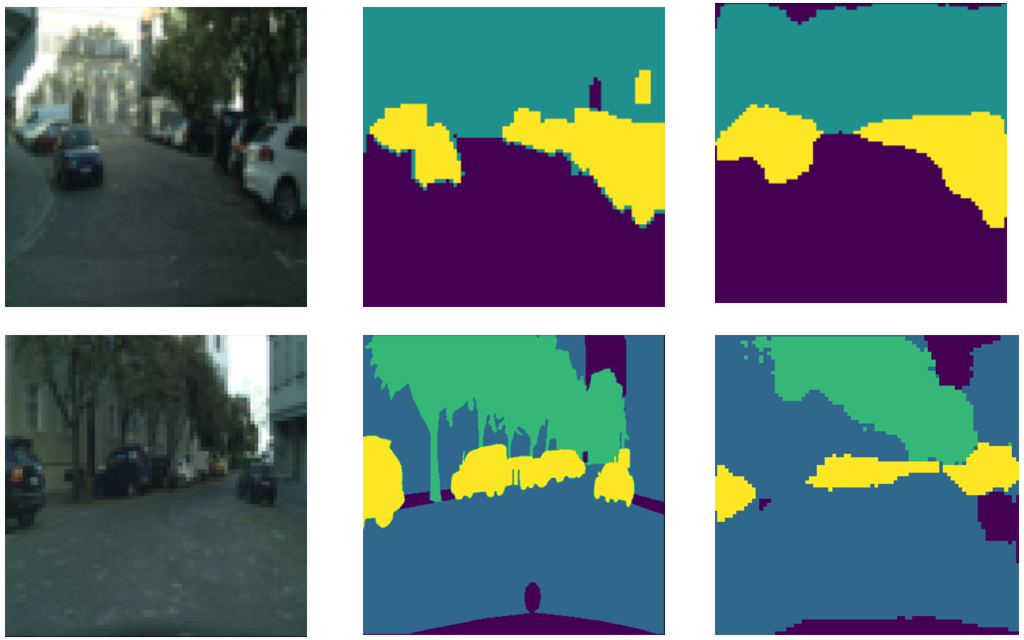
Model Testing: Comparison on Test Images from Cityscapes

STM32 – Enhanced Network



Test Image　Ground Truth　Predicted Label

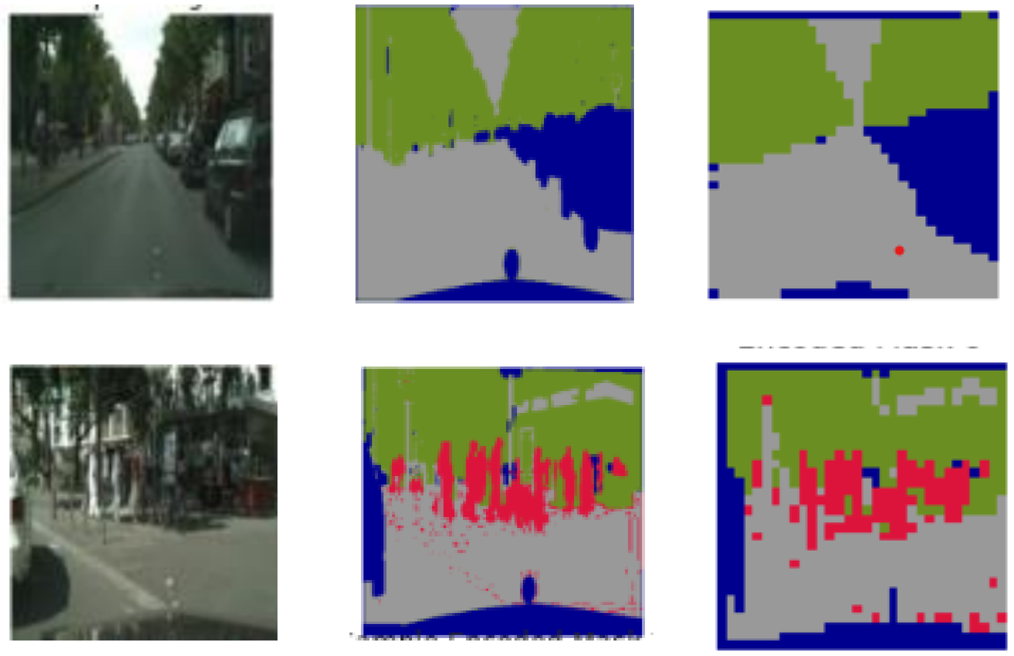
MAX78000 – Enhanced Network



Test Image　Ground Truth　Predicted Label

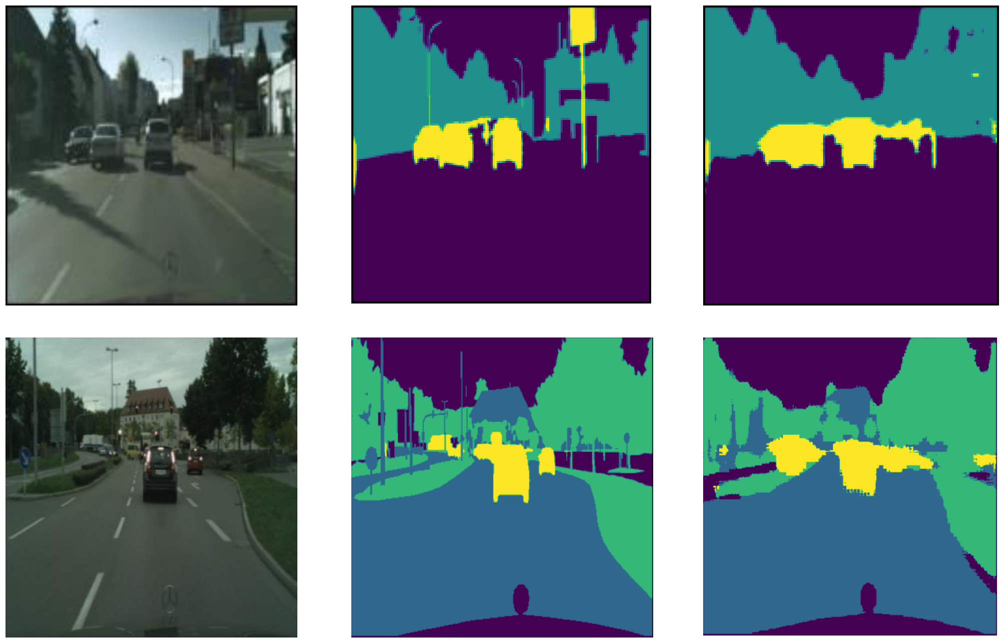
Model Testing: Comparison on Test Images from Cityscapes

STM32 – Small U-Net



Test Image Ground Truth Predicted Label

MAX78000 – Large U-Net



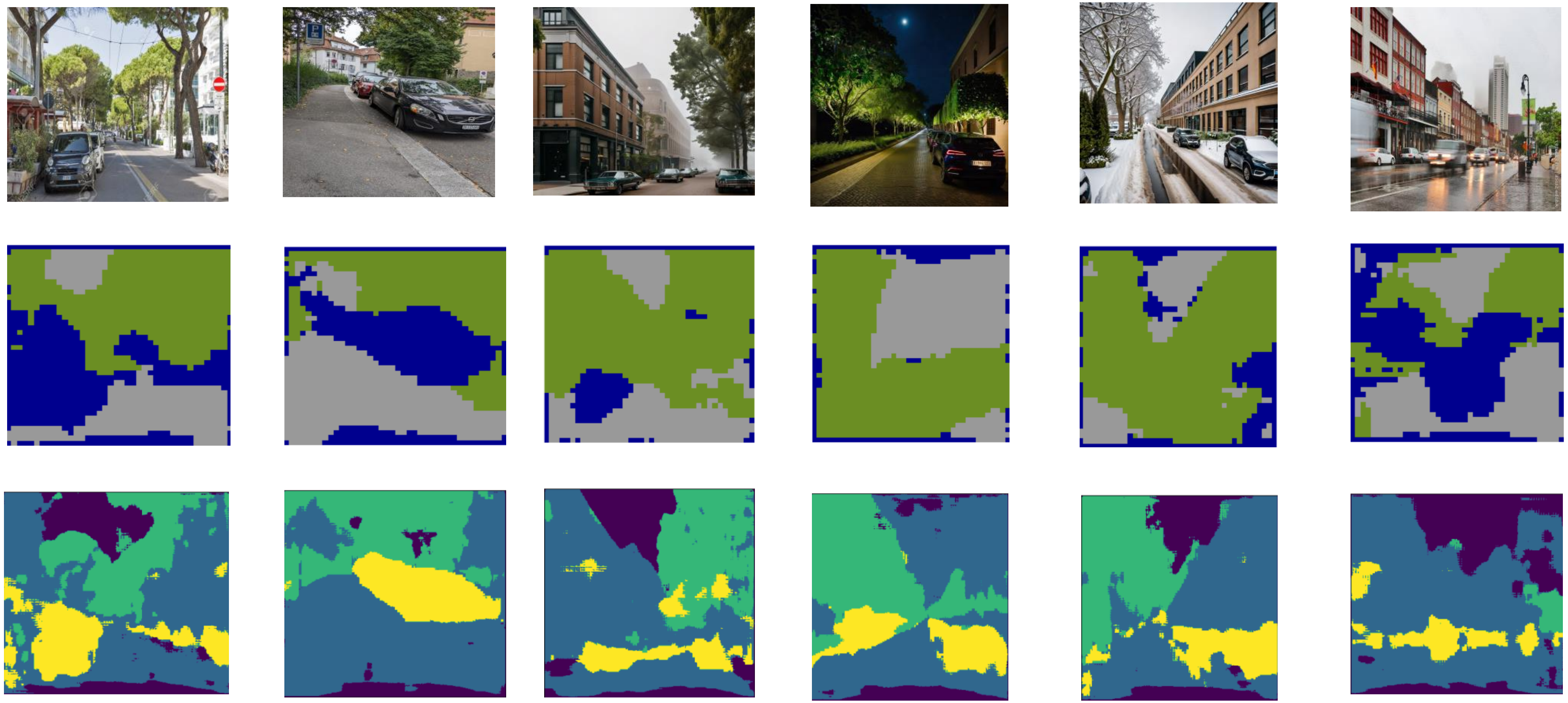
Test Image Ground Truth Predicted Label

Model Testing: Comparison on Images outside Cityscapes

Test Image

STM32M4 Predictions-Enhanced

MAX78000 Predictions - UNet Large



Limitations & Future Work

- The number of supported semantic segmentation classes are limited.
- Cityscapes Dataset contains images that are only captured during the day in good/medium weather conditions.
- Thus, the model performs poorly on night urban traffic images and images with snow, fog, etc.
- The model performs better using float32 in STM32M4 which leads to high memory consumption.
- The inference time must be reduced further for real-time applications such as autonomous driving.
- Transfer Learning techniques and a more broader dataset can be used to improve the model performance.

DEMO

TRAFFIC IMAGE SEGMENTATION ON STM32M4 & MAX78000

MACHINE LEARNING ON MICROCONTROLLERS

ETH ZURICH, JAN 2024

PHANI JAYANTH JONNALAGADDA

VARSHA JAYAPRAKASH